

# Urbanization Research — Case Study 01: Godagama 10200

Date: 2026-06-27

Case-study area: Godagama town, Colombo District, Western Province, Sri Lanka.

Repository: **Urbanization Research**. Godagama is the first city/town case study.

Public repo: <https://github.com/minzique/urbanization-research>

## Purpose

This report models early urbanization pressure around Godagama using multiple evidence layers: Google-derived points of interest, OpenStreetMap history, official UDA/RDA planning documents, expressway/interchange context, developer market signals, and identified open datasets for future satellite/census expansion.

The report is data-led. Recommendations are included only where they follow directly from the measured pattern.

## Executive findings

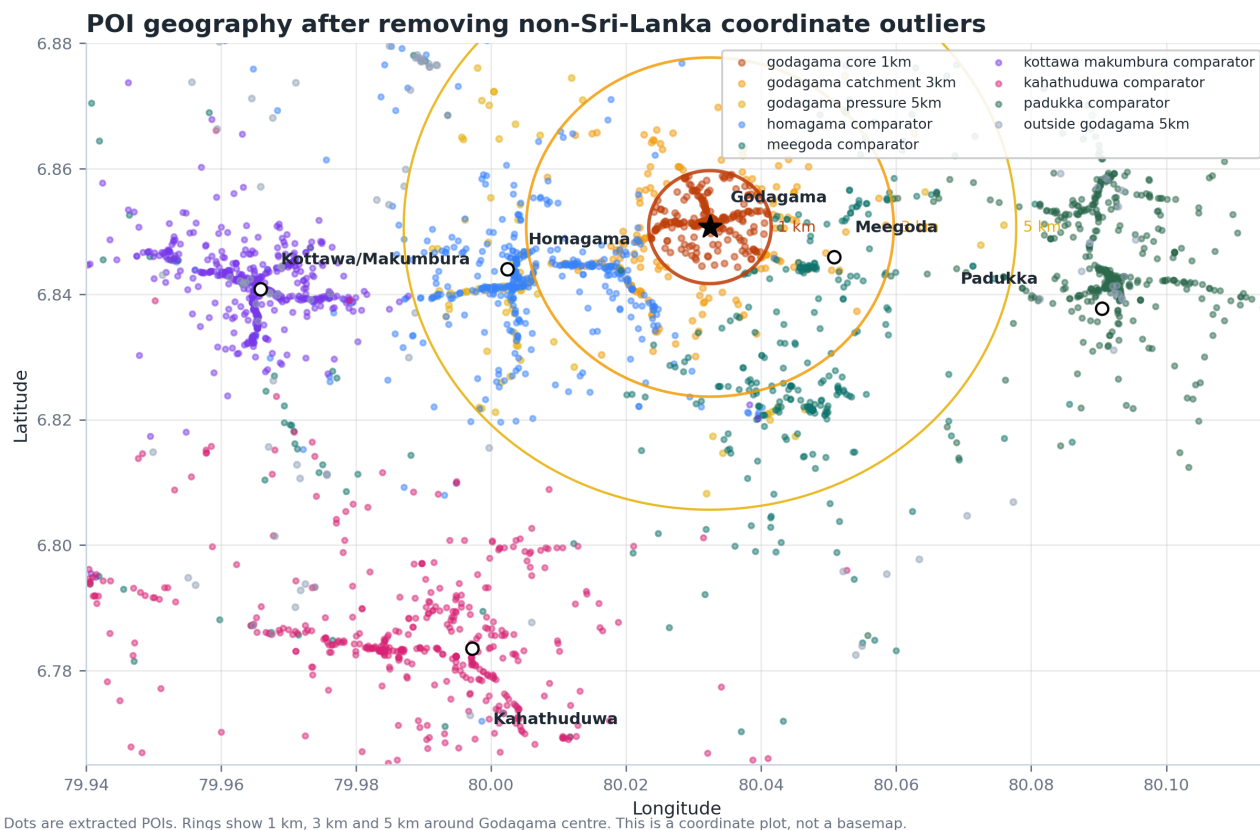
- Godagama has a dense core and a thinner surrounding catchment.** The 3 km catchment has **440 extracted POIs (15.6/km<sup>2</sup>)**. The 1 km core alone has **274 POIs**, about **87.2/km<sup>2</sup>**.
- The current Godagama POI layer is recent.** **82.4%** of dated Godagama POIs were first visible from **2021 onward**, the highest recent-share among the six comparator areas in this extract.
- Property/developer pressure is already visible.** Godagama has **63 property/developer POIs** in the 3 km catchment, **14.3%** of the extracted POI base. That share is close to Homagama and Kottawa/Makumbura.
- Comparator pressure is strongest in Kahathuduwa, then Meegoda and Kottawa/Makumbura.** Current score ranking: **Kahathuduwa (72.2)**, **Meegoda (57.6)**, **Kottawa/Makumbura (53.6)**, **Godagama (34.5)**.
- OSM history shows physical growth before full commercial maturity.** In the Godagama-area OSM bounding box, mapped buildings rose **+338%** from 2018 to 2026, while mapped road length rose **+33%**.

## Evidence layers used

| Layer                      | What it contributes                                                              | Status in this report                                                                           | Main limitation                                                                                                 |
|----------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Google-derived POI extract | Current places, category mix, rough first-visible activity dates, chains/anchors | Quantitative: 3,494 geospatially valid POIs after removing one non-Sri-Lanka coordinate outlier | Earliest photo/activity date is not a verified opening date; review count and first review date not decoded yet |

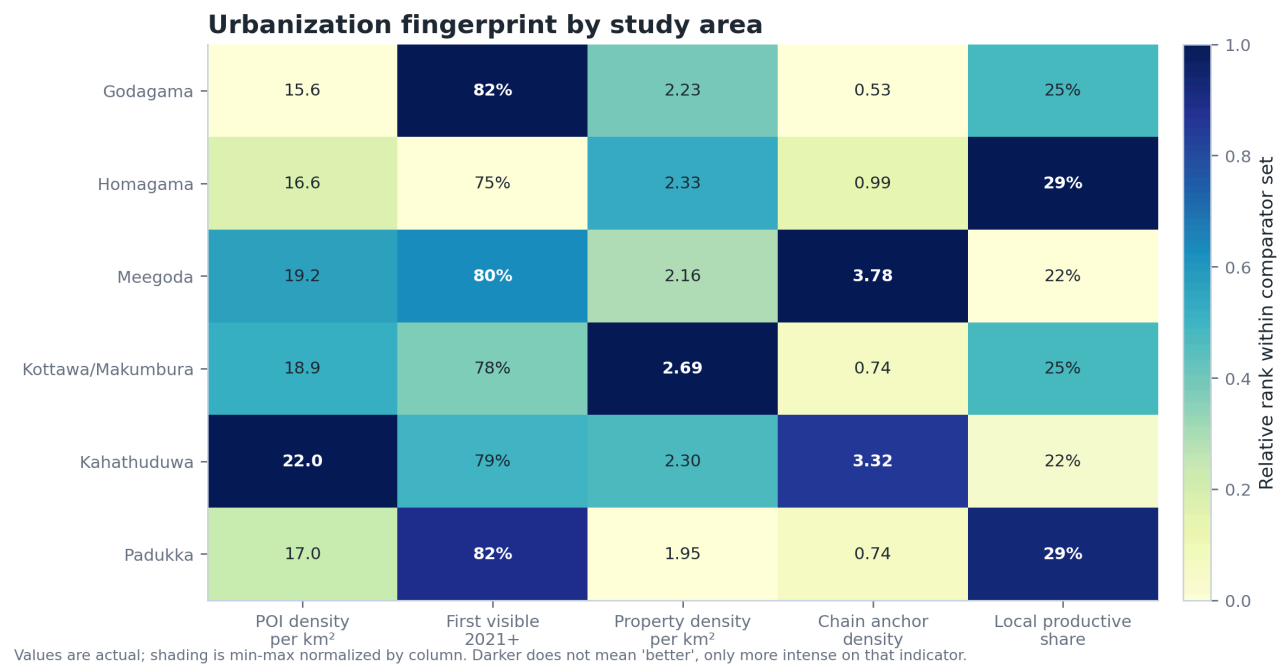
| Layer                                  | What it contributes                                                            | Status in this report                                                                                                       | Main limitation                                             |
|----------------------------------------|--------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| OpenStreetMap / ohsome history         | Mapped buildings, roads, shops, food/drink places over time                    | Quantitative baseline: 2014–2026 time slices, charted as indexed growth                                                     | Affected by mapping completeness and edit bursts            |
| UDA Homagama Development Plan          | Official zoning, town hierarchy, environmental control zones, guide-plan areas | Integrated as planning evidence: Godagama HD Commercial Zone III; Kahathuduwa HD Commercial Zone II; wetland/paddy controls | Needs GIS polygon extraction for parcel-level analysis      |
| RDA / Exway / gazettes                 | Expressway chronology, active interchanges, official corridor status           | Integrated as transport context: OCH, Southern Expressway, Central Expressway comparator nodes                              | Needs exact GIS alignment/interchange layers in the model   |
| Developer market signals               | How land developers market highway access and future value                     | Integrated qualitatively and via Google land-development POIs                                                               | Need full land-price/time-series scrape for price model     |
| DCS Census Portal / district handbooks | Population, housing, labour, local-authority context                           | Source identified; not yet numerically integrated                                                                           | Next pass should export DS/GN-level 2012 vs 2024 fields     |
| GHSL / WorldPop / VIIRS                | Satellite built-up, settlement growth, night-time economic intensity           | Source stack identified for next model expansion                                                                            | Not yet extracted into the Godagama case-study calculations |

## Study geography



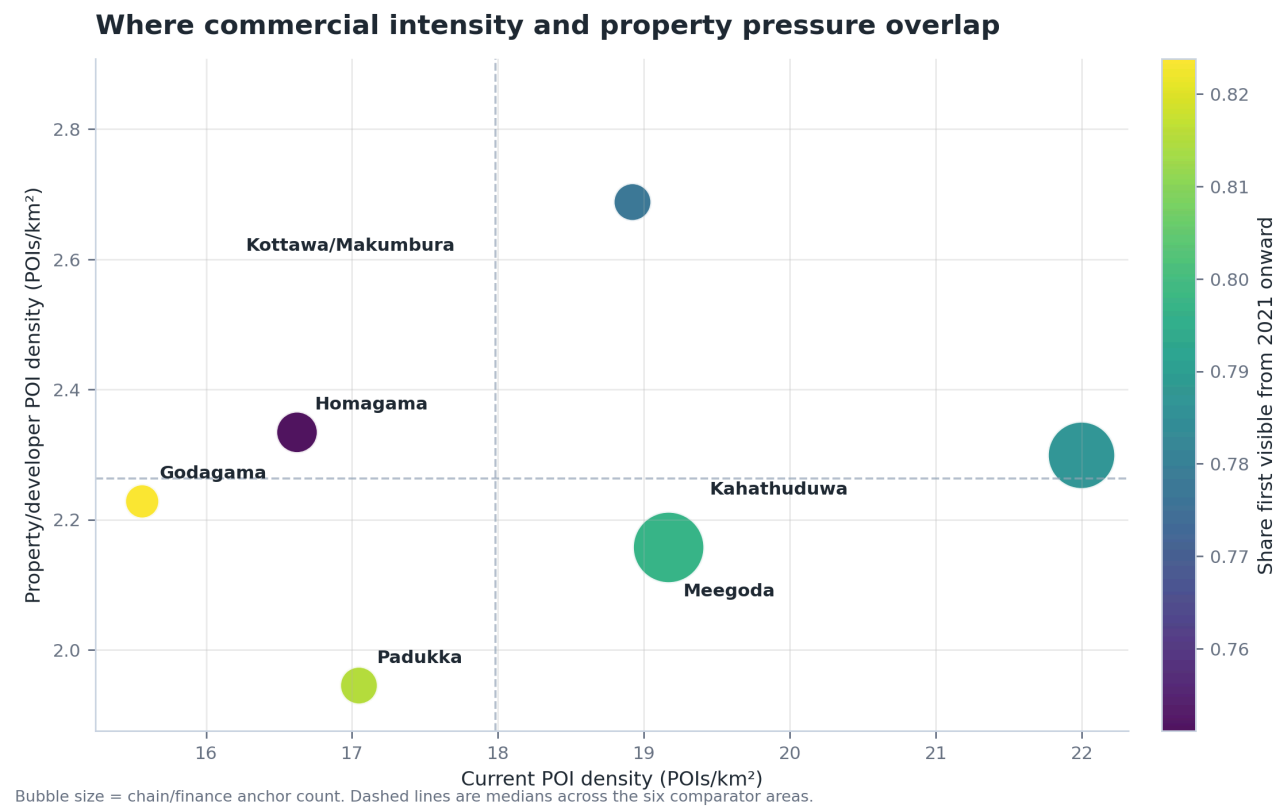
# Urbanization fingerprint

Darker cells show which places are high on each measured indicator; printed values are actual values.



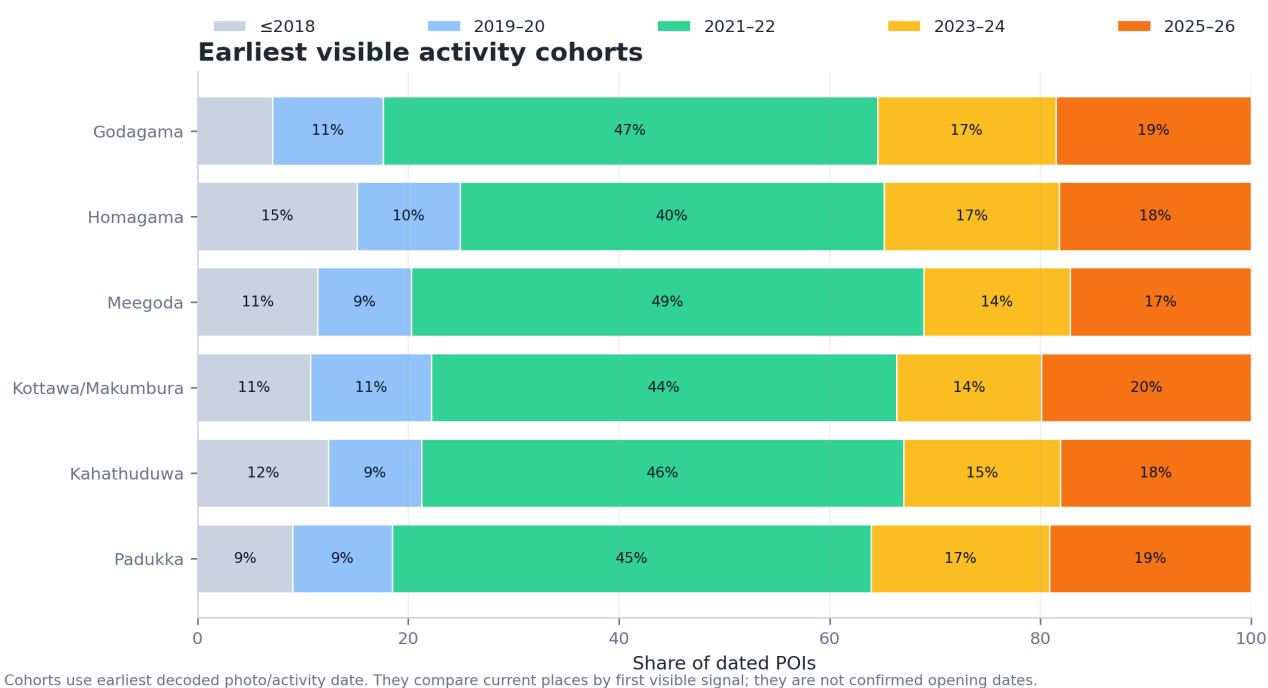
# Pressure model

The score is a triage index, not a land-price forecast. Inputs: 35% POI density, 25% 2021+ first-visible share, 25% property/developer density, and 15% chain/finance anchor density.



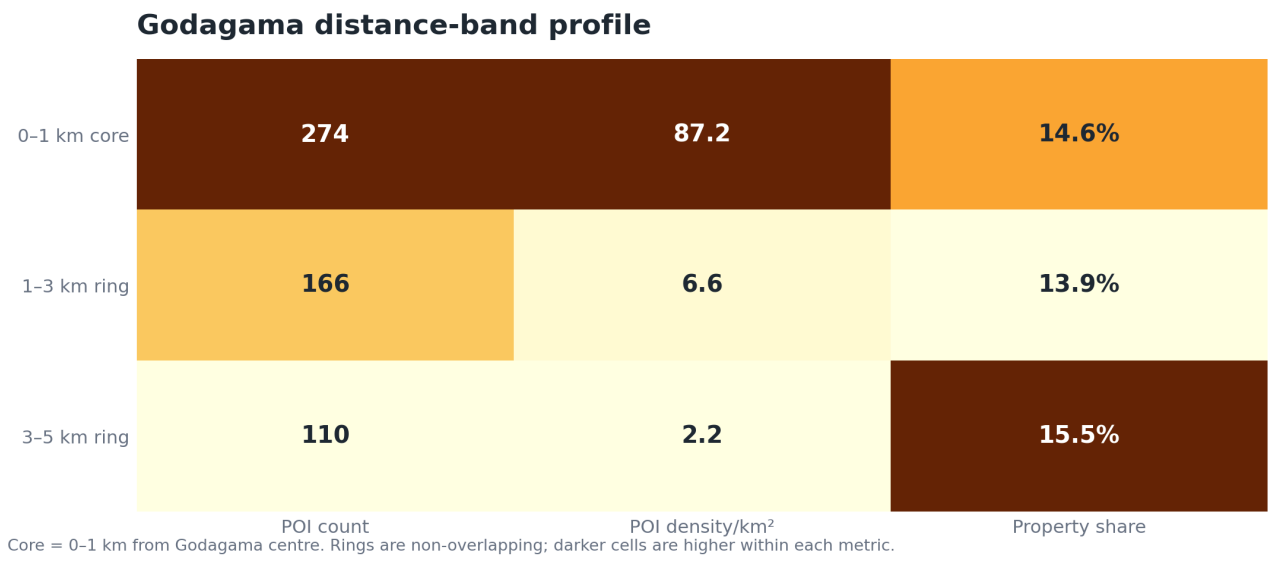
| Area              | POIs | POIs/km <sup>2</sup> | Visible 2021+ | Property share | Property POIs | Productive-service POIs | Score |
|-------------------|------|----------------------|---------------|----------------|---------------|-------------------------|-------|
| Godagama          | 440  | 15.6                 | 82.4%         | 14.3%          | 63            | 111                     | 34.5  |
| Homagama          | 470  | 16.6                 | 75.1%         | 14.0%          | 66            | 137                     | 21.0  |
| Meegoda           | 542  | 19.2                 | 79.7%         | 11.3%          | 61            | 117                     | 57.6  |
| Kottawa/Makumbura | 535  | 18.9                 | 77.8%         | 14.2%          | 76            | 135                     | 53.6  |
| Kahathuduwa       | 622  | 22.0                 | 78.7%         | 10.5%          | 65            | 136                     | 72.2  |
| Padukka           | 482  | 17.0                 | 81.5%         | 11.4%          | 55            | 138                     | 31.2  |

## Timeline signal



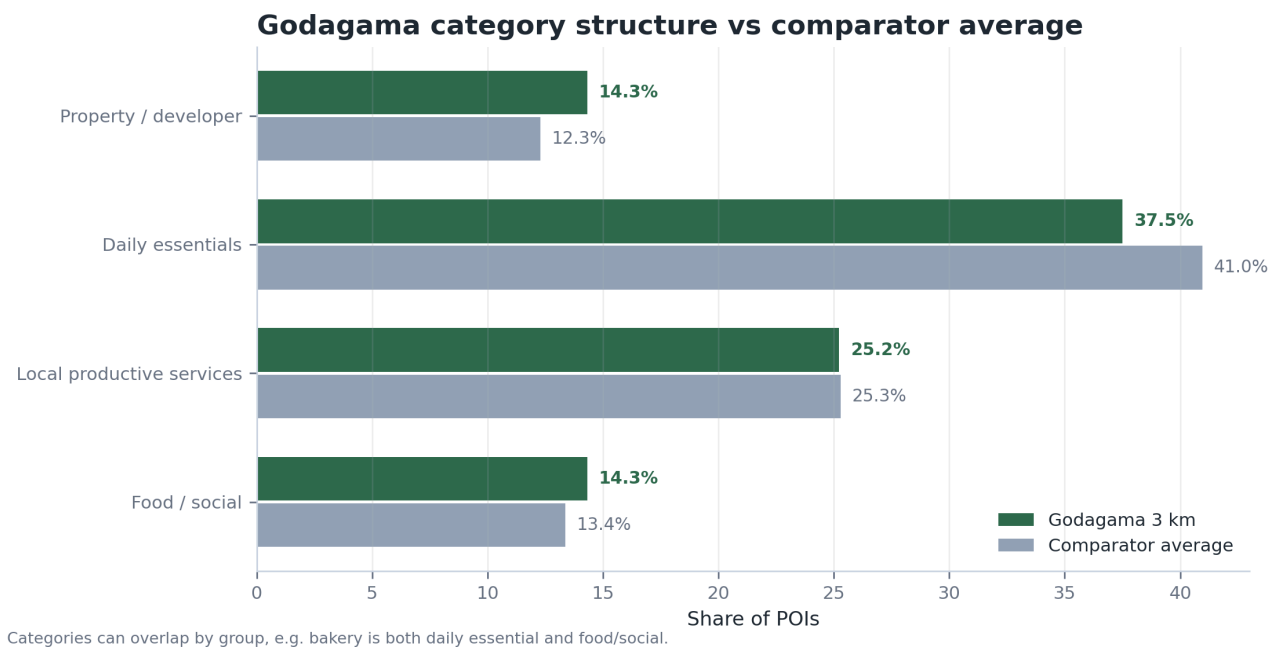
The 2021–2022 band is large across all areas. Treat it as a combined real activity + Google photo coverage signal. The comparison remains useful because the same extraction method is applied to every area.

# Godagama distance-band structure



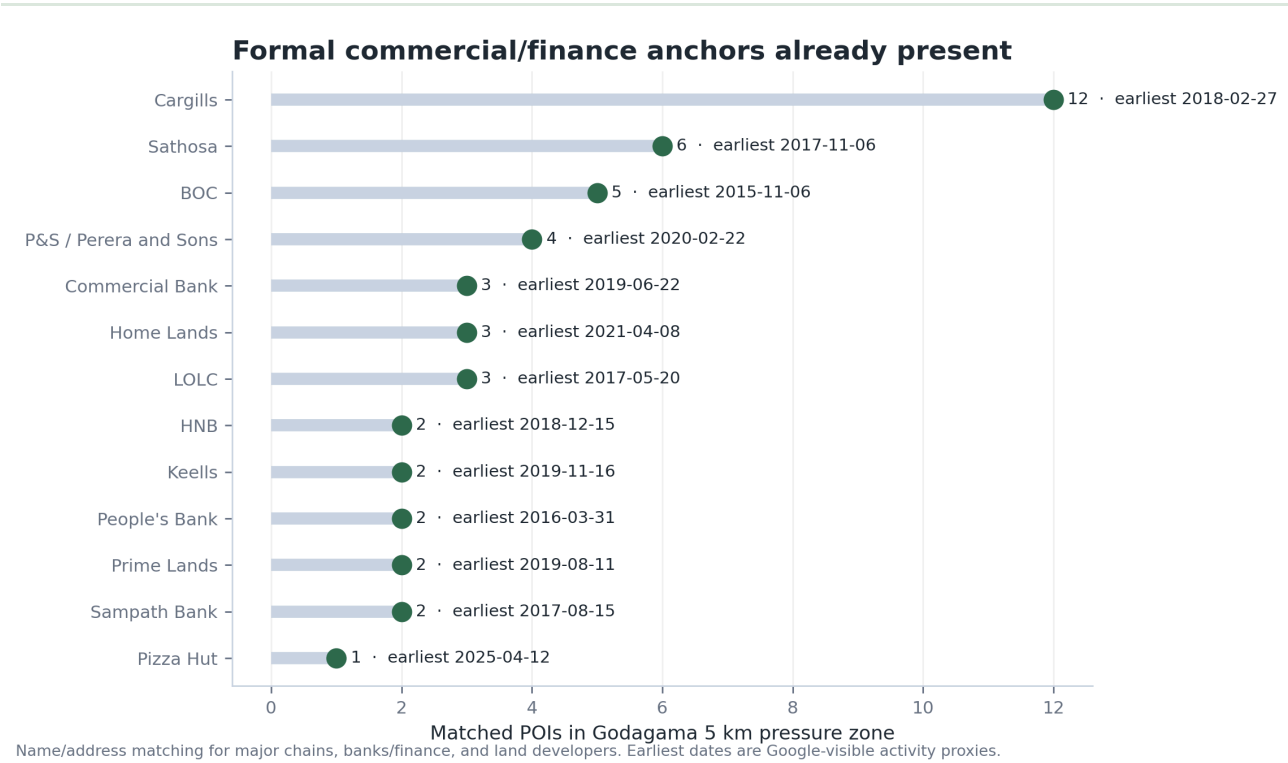
The 1 km core carries most of the visible density. The 1-5 km rings have lower density but still show property/developer pressure, so the town needs both core management and outer-ring growth guidance.

## Category structure



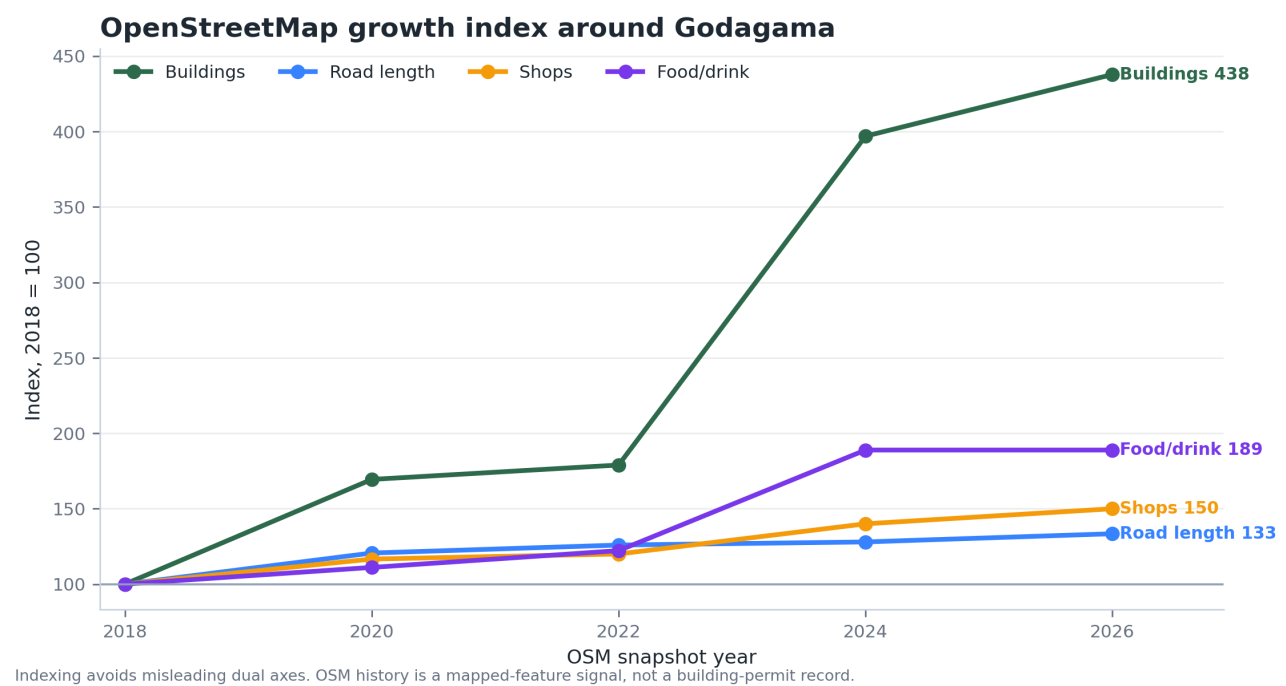
Godagama is not only a food or commuting suburb. The extracted mix shows daily essentials, clinics, hardware, vehicle service, electronics, courier, apartments, and land development. That supports a practical local-enterprise framing rather than a single-sector framing.

# Chain, finance, and developer anchors



| Anchor                | Matched POIs in Godagama 5 km | Earliest visible date |
|-----------------------|-------------------------------|-----------------------|
| Cargills              | 12                            | 2018-02-27            |
| Sathosa               | 6                             | 2017-11-06            |
| BOC                   | 5                             | 2015-11-06            |
| P&S / Perera and Sons | 4                             | 2020-02-22            |
| Commercial Bank       | 3                             | 2019-06-22            |
| Home Lands            | 3                             | 2021-04-08            |
| LOLC                  | 3                             | 2017-05-20            |
| HNB                   | 2                             | 2018-12-15            |
| Keells                | 2                             | 2019-11-16            |
| People’s Bank         | 2                             | 2016-03-31            |
| Prime Lands           | 2                             | 2019-08-11            |
| Sampath Bank          | 2                             | 2017-08-15            |
| Pizza Hut             | 1                             | 2025-04-12            |

# Physical growth baseline from OSM



OpenStreetMap history is not a building-permit record, but it shows the direction and scale of mapped physical growth. The 2018–2026 building and road signals are consistent with subdivision/build-out preceding a mature commercial layer.

## Official planning and expressway context

| Finding                                                          | Evidence collected                                                                                                                                                                     | Planning/model implication                                                              |
|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Godagama is not just passive sprawl                              | UDA Homagama plan identifies Godagama as High-Density Commercial Zone III                                                                                                              | Treat Godagama as an official intensification node                                      |
| Kahathuduwa is a stronger interchange comparison                 | UDA plan identifies Kahathuduwa as High-Density Commercial Zone II and a guide-plan area; Kahathuduwa interchange is within Homagama PS                                                | Use Kahathuduwa as the near-future comparator for interchange-driven growth             |
| Highway access shapes market language                            | Prime Lands markets Dagny Godagama through Makumbura/Athurugiriya access; Landify Homagama through Kottawa hub/highway exit; Home Lands markets projects through interchange proximity | Developer listings are early-warning indicators for land-price and subdivision pressure |
| Environmental constraints are explicit                           | UDA plan includes Wetland Nature Conservation Zone and Paddy/Wetland Agricultural Zone; sensitive commercial projects trigger environmental/planning clearance                         | Map protect / guide / intensify zones before approving outer-ring growth                |
| Central Expressway is a comparator, not the main Godagama driver | CE Phase II nodes such as Mirigama/Kurunegala are useful examples of interchange-led peri-urban land products                                                                          | Use CE nodes to compare how land products appear before full commercial maturity        |

## Developer market signals collected

| Developer / project signal           | Extracted evidence                                                                                                                                                            | Use                                            |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Prime Lands — Dagny Godagama         | Marketed as rapidly developing, near High Level Road, Godagama Junction, Meegoda Economic Centre, Makumbura/Athurugiriya access; advertised at LKR 1,240,000 per perch upward | Direct Godagama land-market pressure signal    |
| Prime Lands — Landify Homagama       | Marketed near Homagama town, Kottawa Multimodal Hub, Kottawa Highway Exit; advertised at LKR 2,250,000 per perch upward                                                       | Comparator price/access signal                 |
| Prime Lands — Ever Green Kahathuduwa | Marketed through Kahathuduwa interchange access                                                                                                                               | Interchange-led comparator signal              |
| Home Lands inventory/API             | Projects collected for Meepe, Kahathuduwa, Athurugiriya/Panagoda, Mirigama with block counts, availability, prices/access claims                                              | Useful for absorption and price-pressure model |

## Data-driven planning suggestions

1. **Core 0–1 km:** manage as a town centre. Priorities: crossings, drainage, frontage rules, parking/loading, shade, and small-shop continuity.
2. **Ring 1–3 km:** reserve room for local productive services: hardware, repair, vehicle service, electronics, courier, clinics, food, bakeries, and plant/garden businesses.
3. **Ring 3–5 km:** require drainage, paddy/wetland screening, road-connectivity review, and developer contribution before large approvals.
4. **Quarterly monitoring:** rerun the POI extraction and compare Godagama against Kahathuduwa and Kottawa/Makumbura.
5. **Next data pass:** add review counts/first-review dates, official opening dates for top anchors, land-price-per-perch data, UDA zoning polygons, GN/DS census fields, and GHSL/WorldPop/VIIRS raster metrics.

## Public repository access

Public repo: <https://github.com/minzique/urbanization-research>

Display title: **Urbanization Research**

First case folder: `cases/godagama-10200/`



Scan for the public repository, source notes, report assets, and case-study data.



Public contents include:

- printable PDF report;
- source markdown report;
- cleaned charts;
- derived summary CSV/JSON and the extracted public POI dataset;
- source notes, methodology, and reproducibility scripts;
- no private session files, raw browser captures, WhatsApp data, or unrelated private project materials.

## Appendix: plain-language terms

| Term in this report     | Plain English                                                                                               | Sinhala term used in the Sinhala version |
|-------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------|
| Urbanization            | A place becoming more town-like: more buildings, shops, services, traffic, and land subdivision             | නාගරීකරණය / නගර ගතිය වැඩි වීම            |
| POI / point of interest | A mapped place such as a shop, bank, clinic, school, fuel station, restaurant, or service business          | සිතියමේ ලකුණු වූ ස්ථානය / සේවා ස්ථානය    |
| Catchment               | The practical area around a town that uses its shops, services, roads, and jobs                             | අවට සේවා පරාසය                           |
| Core                    | The centre of the town; in this report, the 1 km area around Godagama junction/station                      | මධ්‍ය ප්‍රදේශය                           |
| Pressure zone           | An area where new land sales, shops, roads, and services show development pressure                          | වර්ධන පීඩන කලාපය                         |
| First visible date      | The earliest date visible in the extracted online activity/photo signal. It is not a confirmed opening date | මුලින්ම දත්තවල පෙනුණු දිනය               |
| Comparator area         | A nearby place used for comparison, such as Kahathuduwa or Kottawa/Makumbura                                | සැසඳුම් ප්‍රදේශය                         |
| OSM / OpenStreetMap     | A public map database edited by contributors; useful for road/building history, but not a permit record     | OpenStreetMap / විවෘත සිතියම් දත්ත       |
| ohsome                  | A service that summarizes OpenStreetMap history over time                                                   | OSM ඉතිහාස දත්ත සේවාව                    |
| UDA                     | Urban Development Authority; Sri Lanka's official urban planning authority                                  | නාගරික සංවර්ධන අධිකාරිය                  |
| RDA                     | Road Development Authority; Sri Lanka's main road authority                                                 | මාර්ග සංවර්ධන අධිකාරිය                   |
| GN / DS                 | Grama Niladhari and Divisional Secretariat administrative levels                                            | ග්‍රාම නිලධාරී / ප්‍රාදේශීය ලේකම් මට්ටම් |

| Term in this report     | Plain English                                                                                                                         | Sinhala term used in the Sinhala version |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| GHSL / WorldPop / VIIRS | Global datasets for built-up land, population distribution, and night-time lights                                                     | ගෝලීය ජනාවාස / ජනගහන / රාත්‍රී ආලෝක දත්ත |
| Chain anchor            | A known chain shop, bank, supermarket, or finance branch that signals stronger town services                                          | ප්‍රධාන ජාල වෙළඳ/මුදල් සේවා ස්ථානය       |
| Productive-service POIs | Everyday local services that support households and small businesses: hardware, repair, vehicle service, clinics, courier, food, etc. | දෛනික අවශ්‍යතා සපයන සේවා ස්ථාන           |
| Land-price-per-perch    | Land price quoted for one perch, the common Sri Lankan land-sales unit                                                                | පර්ච්පයක ඉඩම් මිල                        |